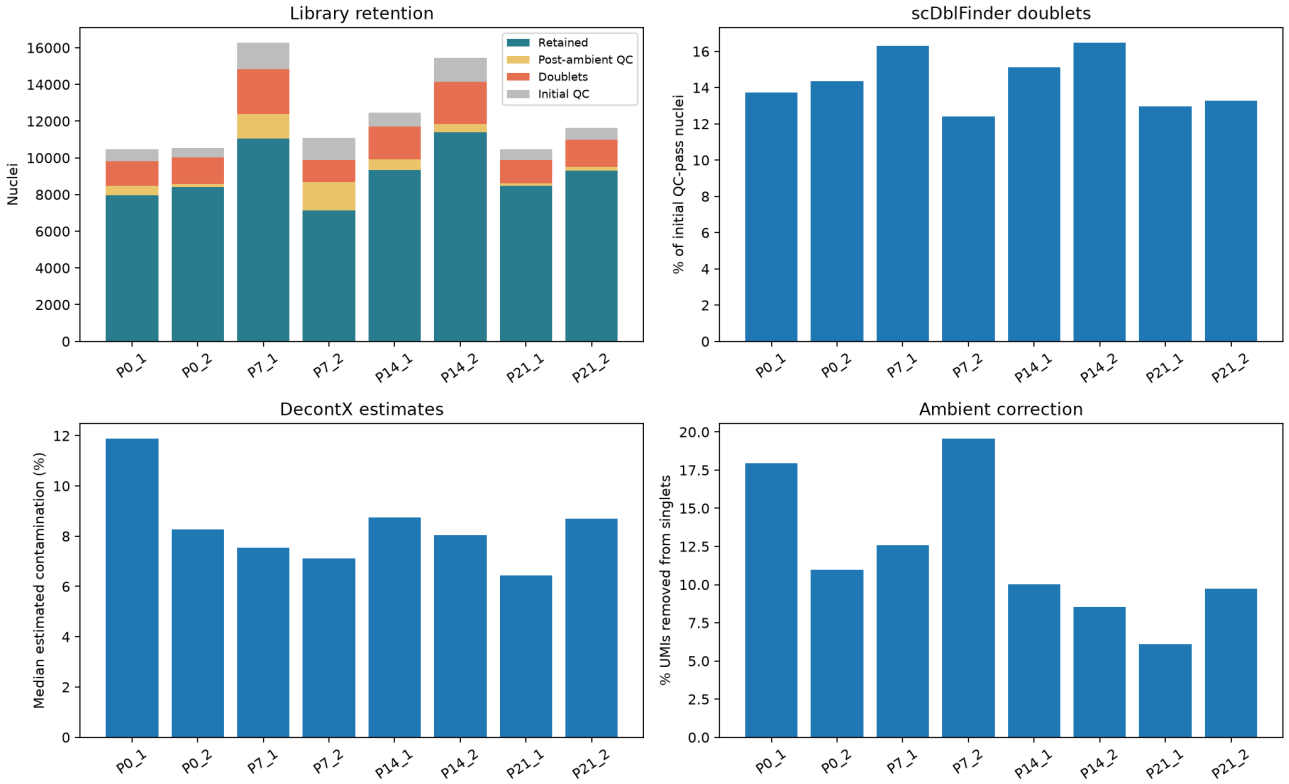
GSE290989: snRNA-seq QC

Processed all eight libraries: 98,371 input barcodes; 73,052 retained nuclei (74.3%).

Initial QC removed 7,090; scDblFinder removed 13,272 predicted doublets from QC-pass nuclei; post-correction QC removed another 4,957. These sequential counts do not overlap.



Library	Input	Doublets	Final	Ambient median
P0_1	10,461	1,349	7,959	11.9%
P0_2	10,535	1,437	8,398	8.3%
P7_1	16,280	2,414	11,043	7.5%
P7_2	11,075	1,228	7,120	7.1%
P14_1	12,467	1,769	9,343	8.8%
P14_2	15,460	2,334	11,380	8.0%
P21_1	10,462	1,283	8,492	6.4%
P21_2	11,631	1,458	9,317	8.7%

Methods and decisions

Each library was processed separately. Original downloaded files remain unchanged. The reference features use mm10 gene IDs with mouse gene symbols. Initial QC retains nuclei with 750-7,500 detected genes, at least 1,000 UMIs, and at most 5% mitochondrial UMIs. The gene and mitochondrial limits reproduce the paper; the minimum UMI threshold is an additional conservative rule. Ribosomal and hemoglobin fractions and within-library three-MAD count/gene flags are recorded but are not automatic exclusions, because cell types and developmental stages differ in RNA content.

Preliminary per-library clusters: counts normalized to 10,000 UMIs, log1p, 2,500 variable genes, 30 PCs, 15-neighbor graph, Leiden resolution 0.6, seed 42. Clusters are computational inputs to the QC models, not final cell-type annotations. No age/batch integration is used for QC.

Primary doublet detection: scDblFinder 1.24.10, supplied preliminary clusters, separate fit per library, default library-size-dependent prior (dbr.per1k=0.008), default learned threshold, serial execution, seed 42. Loading and capture-channel details were unavailable; if a library combines multiple captures, refit using capture-specific IDs. Scrublet scores and automatic calls are diagnostic only. Scrublet returned overly high automatic thresholds and zero calls in some libraries; using that alone would not adequately establish doublet removal. The primary classifier is not forced to remove a fixed percentage.

Ambient correction: DecontX 1.8.0 on predicted singlets, separately per library, with supplied preliminary clusters, default learned Dirichlet prior (estimateDelta=TRUE), maxIter=500, convergence tolerance 0.001, seed 12345. GEO supplies called-nucleus matrices rather than a full empty-droplet background. Therefore this is cluster-based contamination inference, not a SoupX/CellBender background fit. Estimated contamination is uncertain, especially for closely related populations.

Post-correction QC reapplies at least 750 nonzero genes, at least 1,000 corrected UMIs and at most 5% mitochondrial counts. Nuclei with DecontX estimated contamination greater than 50% are additionally excluded as having predominantly estimated ambient signal. Estimates greater than 30% are flagged. Pre-final corrected singlets are saved so the 50% threshold and other post-correction rules can be revisited.

Validation and limits

Checks include original gzip integrity, nonnegative finite corrected counts, unchanged matrix dimensions and gene ordering, unique cell IDs, exact correspondence between scDblFinder singlets and DecontX outputs, agreement of matrix UMI sums with R output, corrected totals no greater than raw totals, and complete accounting of all original barcodes. ZIP/metadata checks from Xenium acquisition are unrelated to this snRNA-seq QC.

Doublet calls and contamination levels are model estimates, not ground truth. Homotypic doublets may remain. DecontX can remove shared biological signal or miss contamination shared by similar populations. Marker profiles are supplied for reviewing overcorrection; no claim of complete ambient removal is made. QC clusters are not biological annotations, and differential expression or developmental conclusions have not been computed in this QC step. Original integer counts are retained; downstream software requiring integer counts must not silently receive fractional corrected counts.

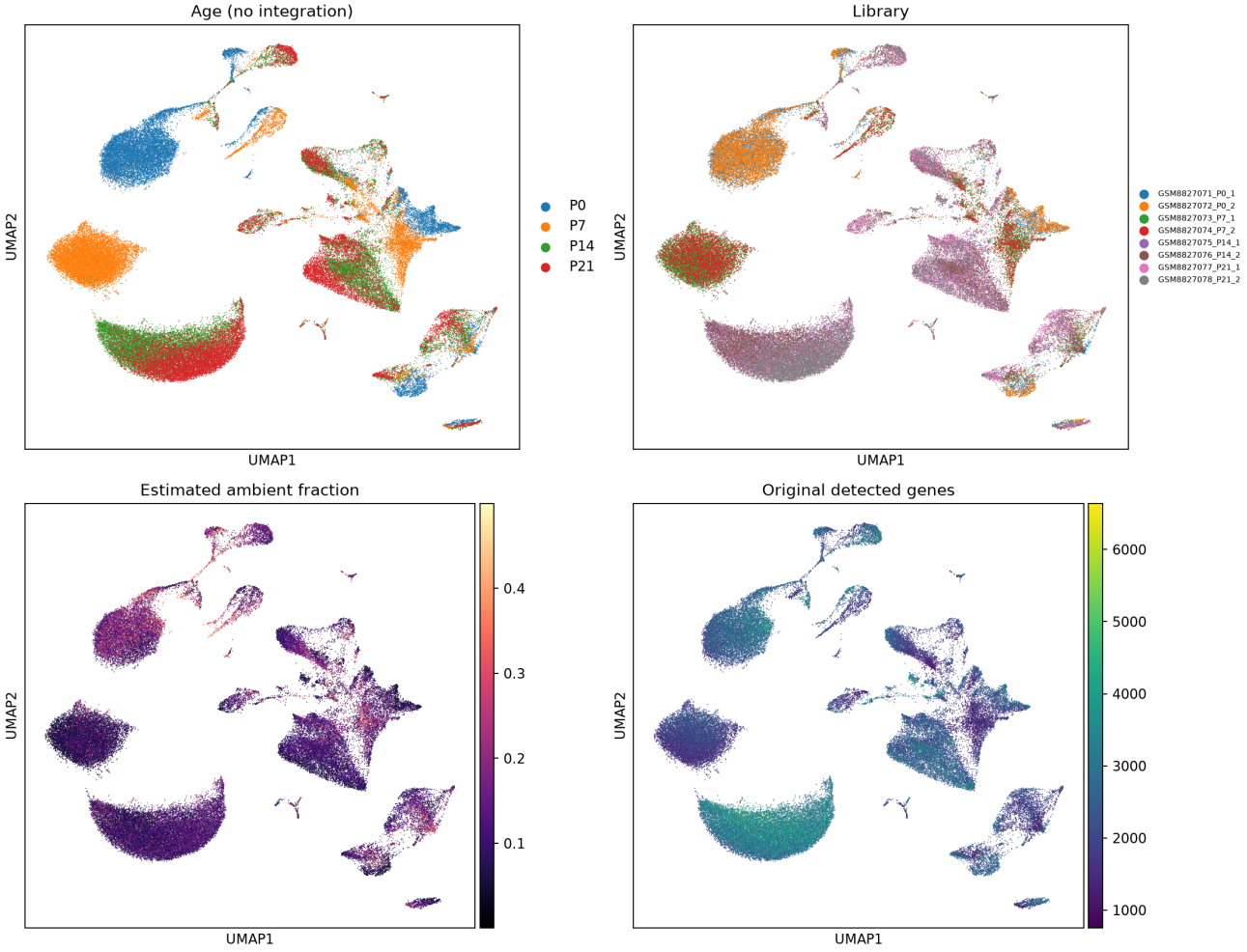
Filtering rates should be considered by age and cell type during later annotation, particularly for rare/low-RNA populations. The paper reports about 67,000 final nuclei; this independent QC is not forced to reproduce that count, and the authors' exact barcode-level exclusions and annotations were not supplied. Alignment quality, intronic/exonic fractions and empty-droplet calling cannot be

reassessed from these called-nucleus count matrices alone. Two libraries per age do not establish animal-level replication; pooling and animal identifiers require confirmation before inferential analyses.

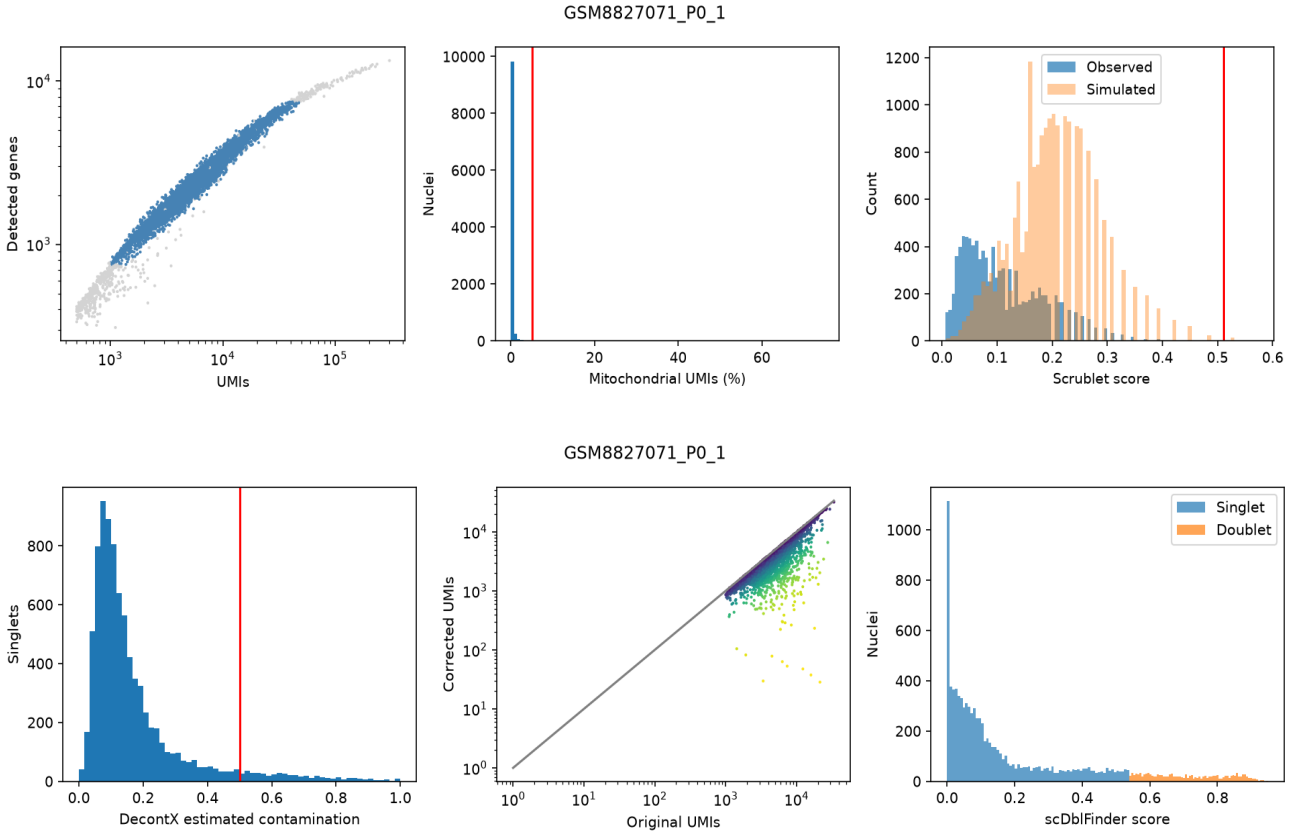
All eight DecontX fits met the convergence tolerance. Per-library iteration counts and final convergence deltas are recorded in `qc_summary.tsv`.

Exploratory QC embedding

No batch integration or age regression. Colors show age, library, fitted contamination and original detected genes. Positions are not final cell-type annotations.

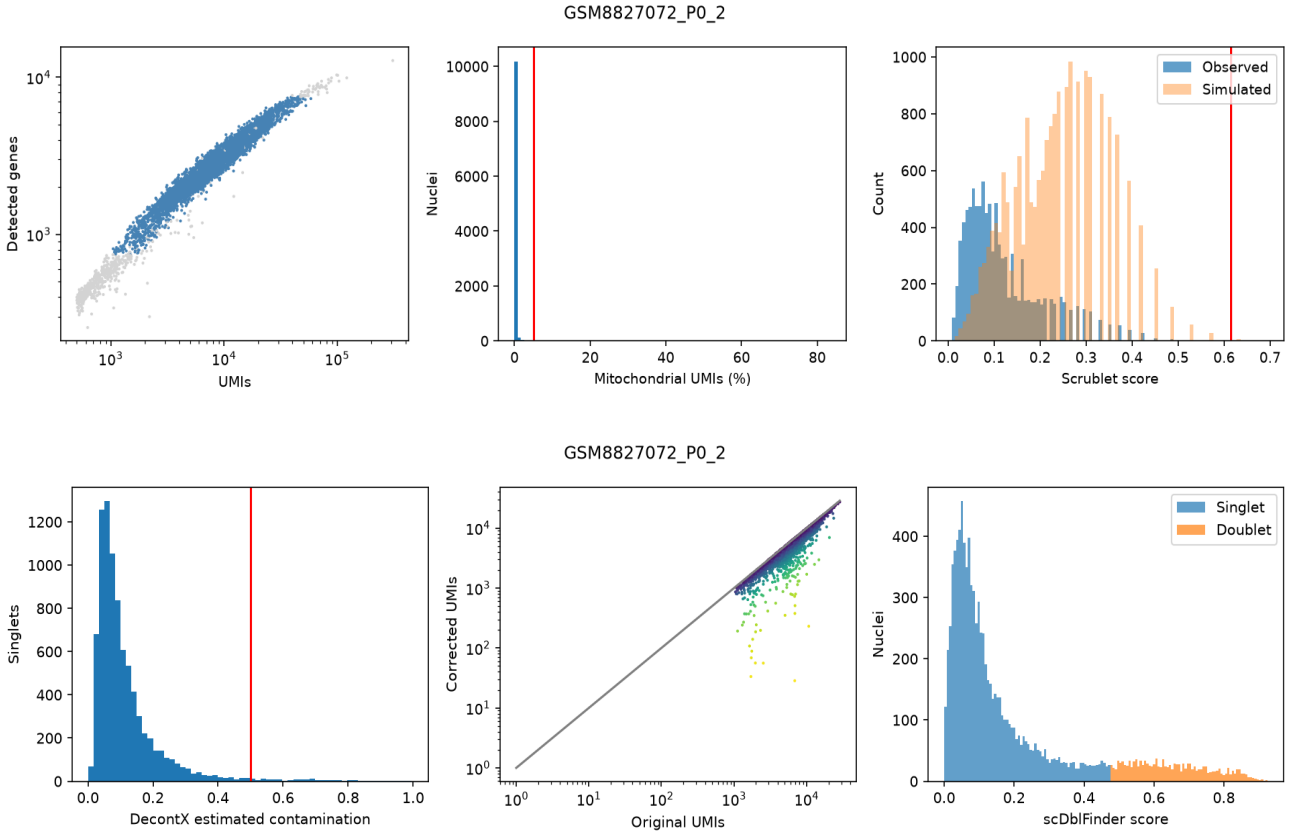


GSM8827071_P0_1



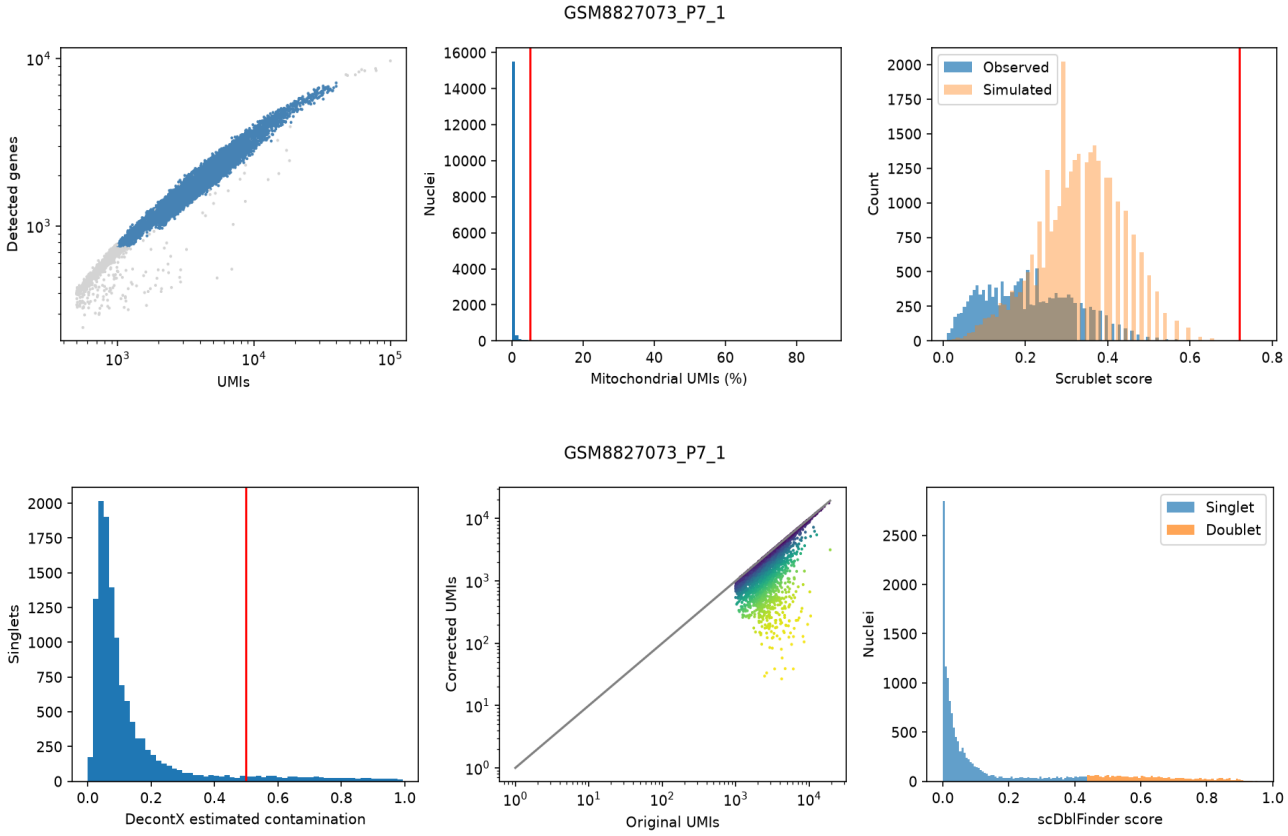
Final nuclei: 7,959; predicted doublets removed: 1,349; median DecontX estimate: 11.9%.

GSM8827072_P0_2



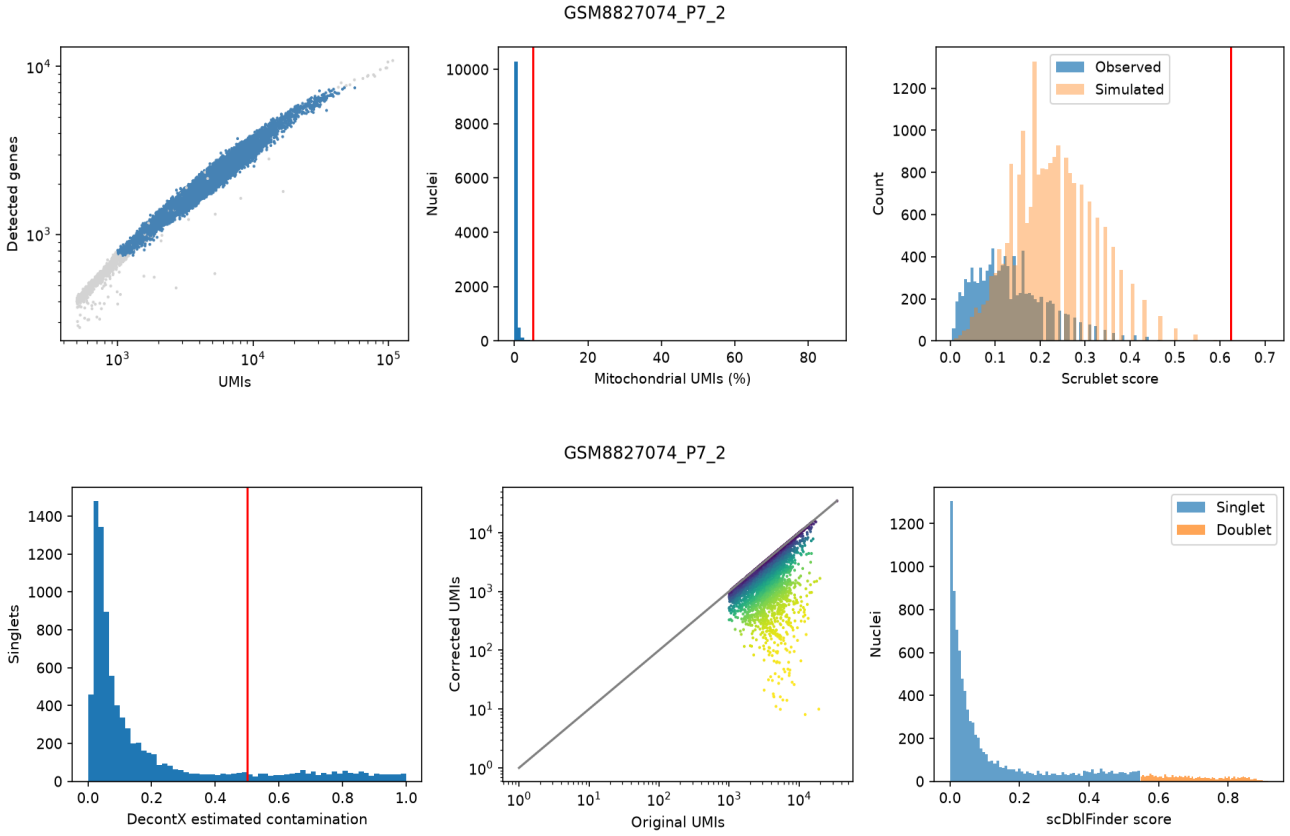
Final nuclei: 8,398; predicted doublets removed: 1,437; median DecontX estimate: 8.3%.

GSM8827073_P7_1



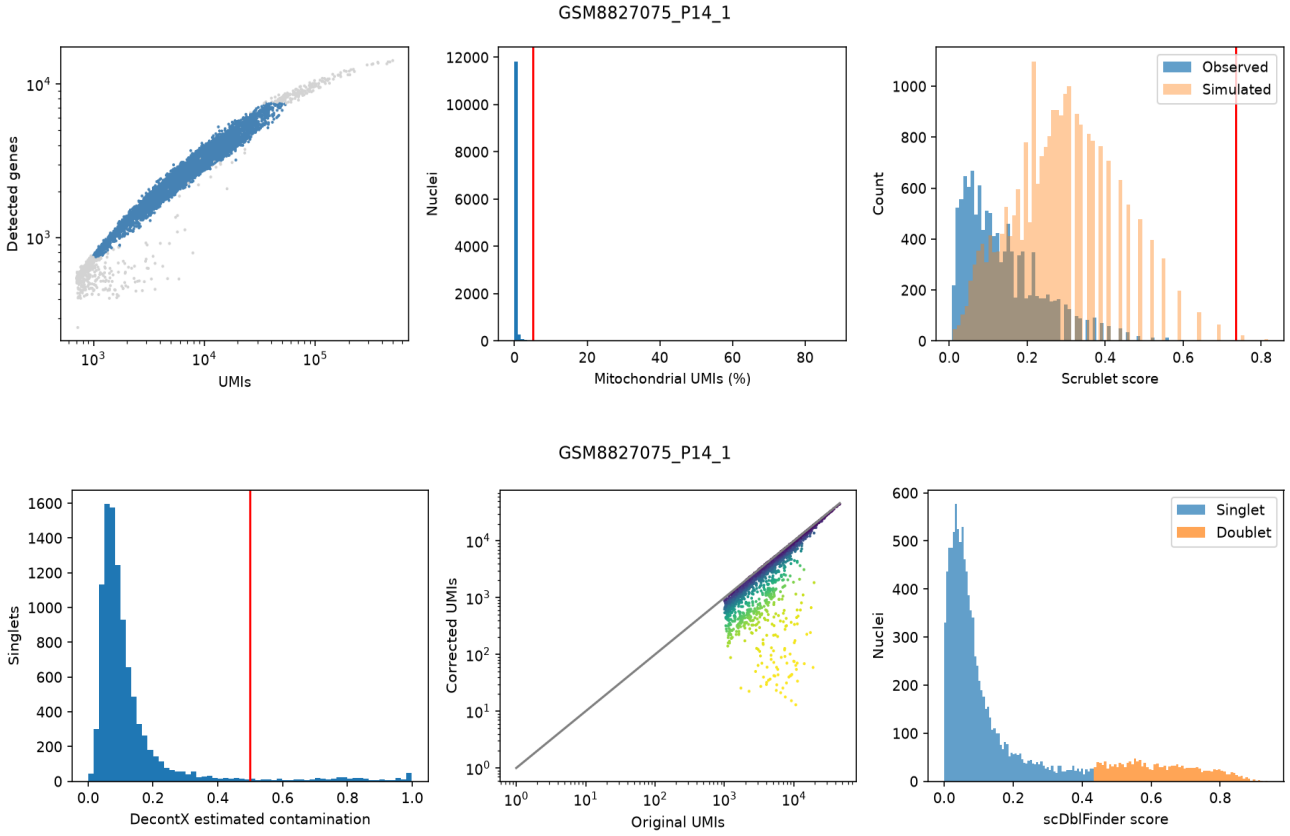
Final nuclei: 11,043; predicted doublets removed: 2,414; median DecontX estimate: 7.5%.

GSM8827074_P7_2



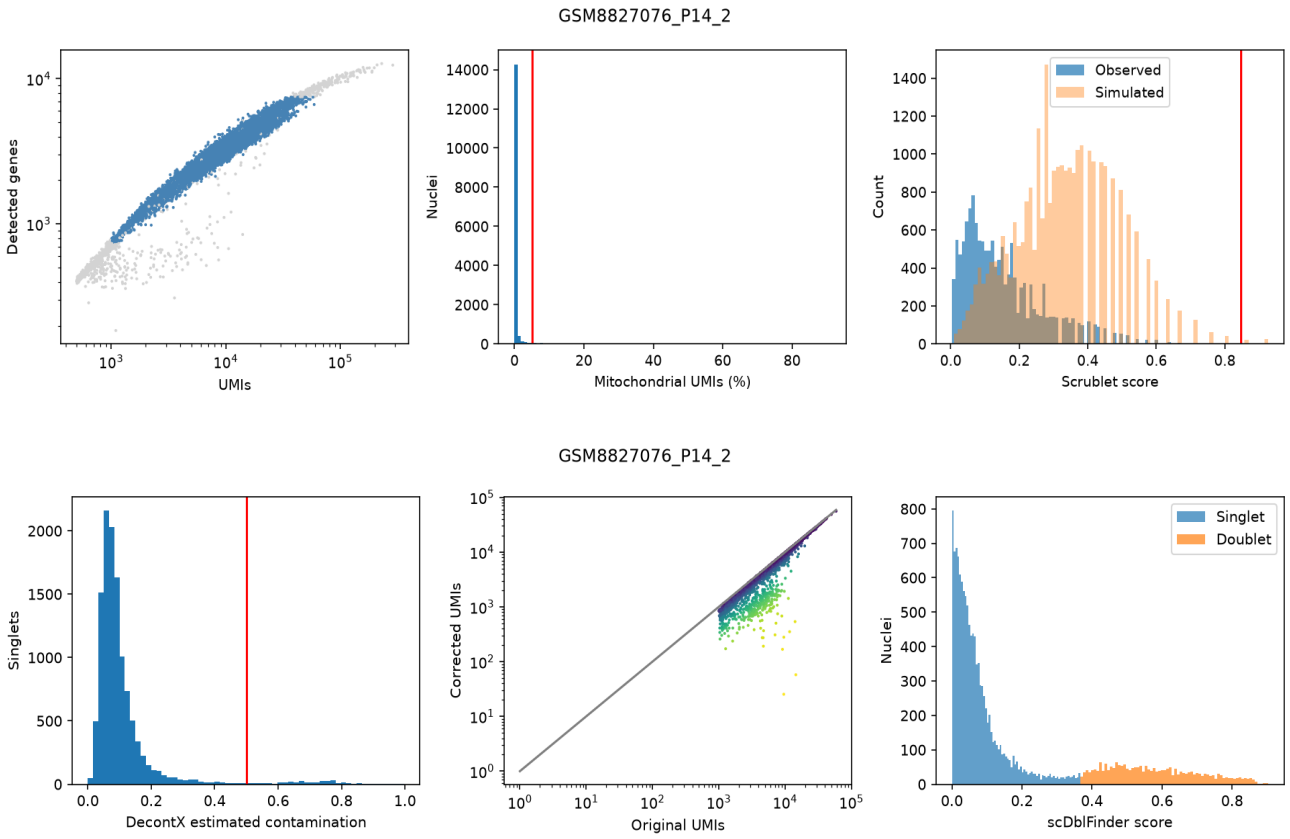
Final nuclei: 7,120; predicted doublets removed: 1,228; median DecontX estimate: 7.1%.

GSM8827075_P14_1



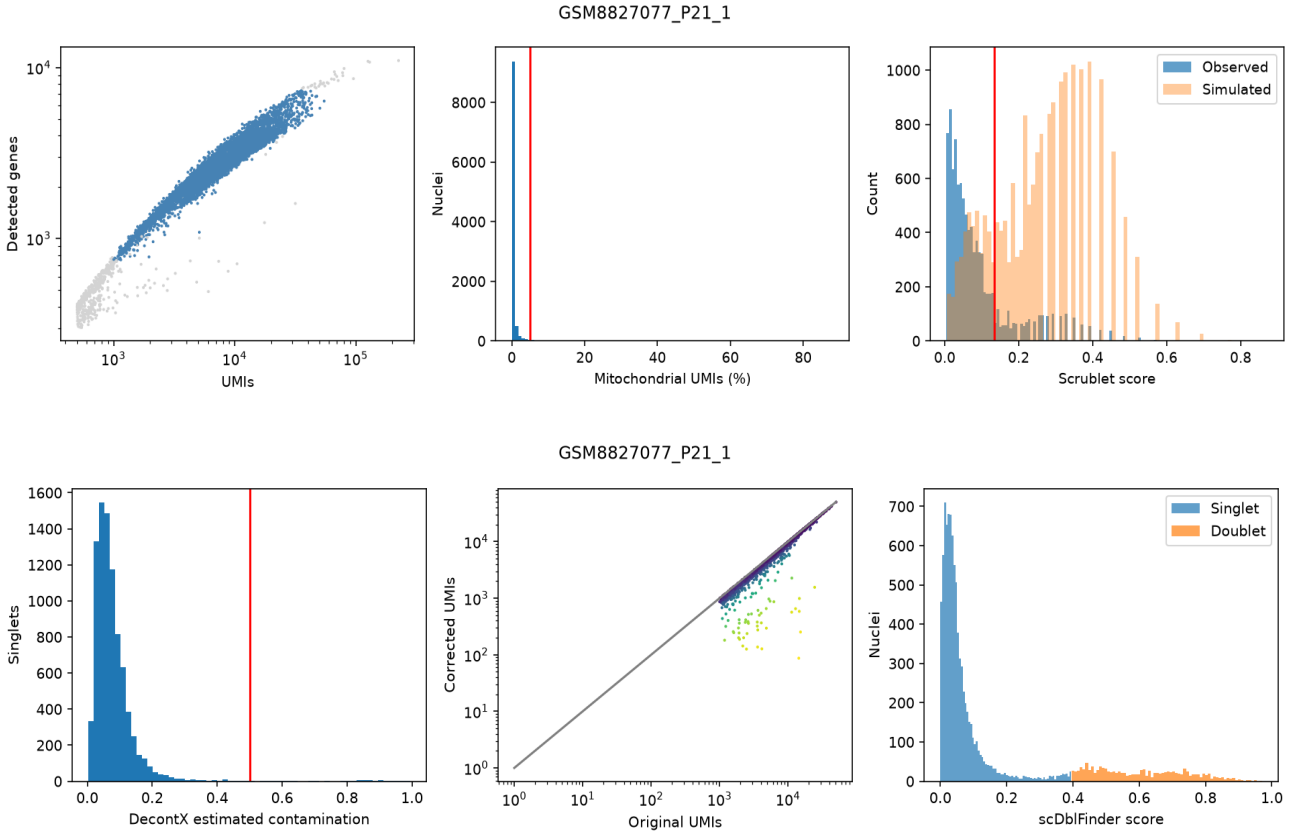
Final nuclei: 9,343; predicted doublets removed: 1,769; median DecontX estimate: 8.8%.

GSM8827076_P14_2



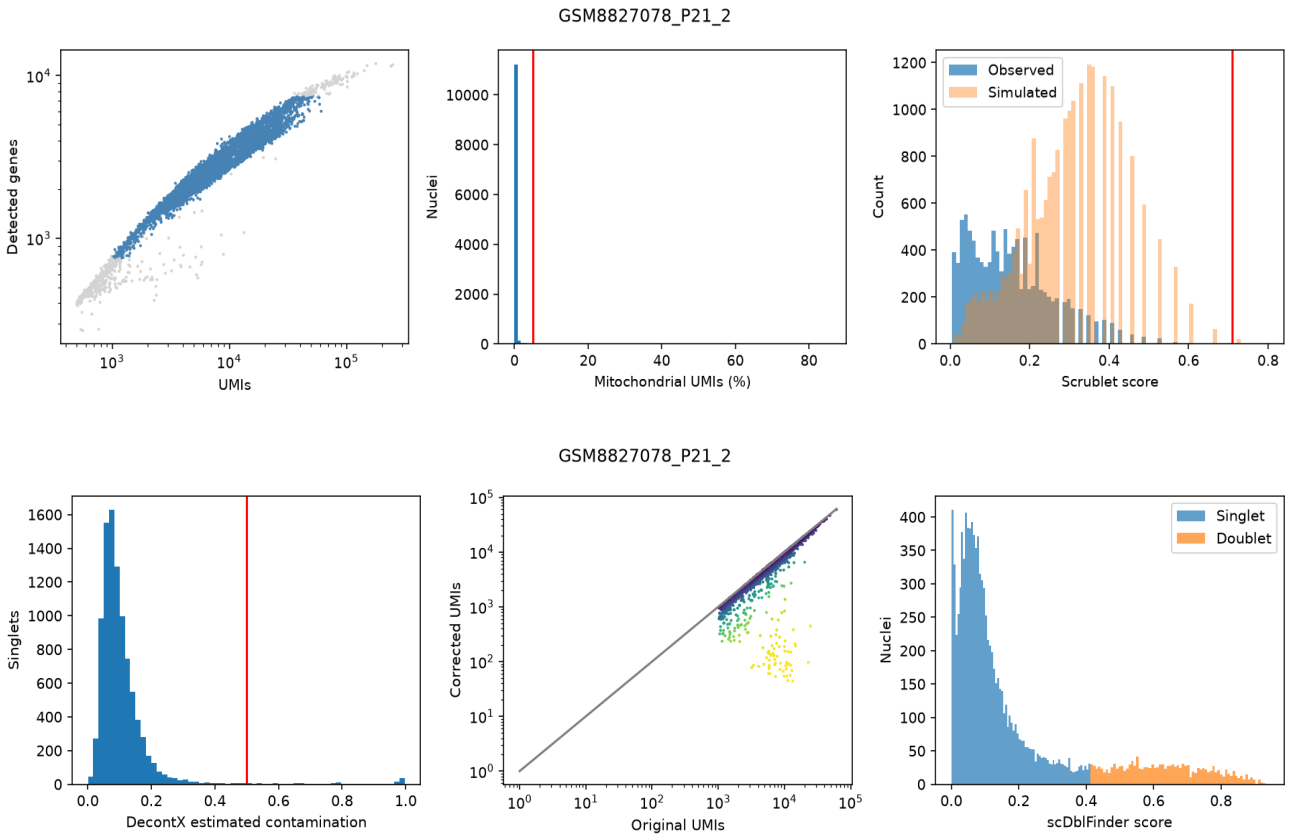
Final nuclei: 11,380; predicted doublets removed: 2,334; median DecontX estimate: 8.0%.

GSM8827077_P21_1



Final nuclei: 8,492; predicted doublets removed: 1,283; median DecontX estimate: 6.4%.

GSM8827078_P21_2



Final nuclei: 9,317; predicted doublets removed: 1,458; median DecontX estimate: 8.7%.